NIDHI SIRIPURAM

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PROFESSIONAL SUMMARY

Data Scientist with **3+ years of experience applying statistical analysis, machine learning, and predictive modeling** to solve complex business problems. Strong ability to act as an **internal consultant**, translating ambiguous business questions into data-driven solutions that support decision-making and process automation. Experienced in **Python, statistics, time series concepts, NLP, distributed data processing**, and communicating insights to technical and non-technical stakeholders.

CORE SKILLS & TECHNICAL EXPERTISE

Programming & Databases: Python, R, SQL, Oracle, MS SQL Server, MongoDB, Cassandra
Machine Learning & AI: Linear & Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, Clustering, Scikit Learn, Tensorflow, Keras
Analytics & Data Science: ARIMA/ARIMAX,Supervised & Unsupervised Learning, NLP, Statistics
Foundation Models & GenAI: Large Language Models (LLMs), Prompt Engineering, Fine-tuning, Retrieval-Augmented Generation (RAG), Transformers, Hugging Face, LangChain
Model Evaluation: RMSE, R^2, Accuracy, Precision, Recall, ROC-AUC, Bias Assessment, Hallucination Detection, Statistical Significance Testing
MLOps & LLMOps: CI/CD Pipelines, Model Monitoring, Retraining Workflows, PromptFlow, Production Deployment
Cloud & Big Data: AWS (EC2, EMR, Bedrock), Azure Machine Learning, Spark/Hadoop MapReduce
Visualization: Tableau, Power BI, matplotlib
Collaboration: Agile/Scrum, Cross-functional Teamwork, Stakeholder Communication

PROFESSIONAL EXPERIENCE

Data Scientist | T-Mobile, USA
May 2024 – Nov 2025

- Worked as an **internal data science consultant**, partnering with business teams to define analytical approaches for complex customer and operational problems.
- Prepared and engineered features from **large-scale structured and unstructured datasets**, improving predictive model performance by **18–22%**.
- Applied **statistical analysis, machine learning, and exploratory data analysis** to uncover drivers of customer behavior.
- Developed and evaluated predictive models using **regression and tree-based methods**, ensuring reliability through robust validation metrics.
- Monitored model performance and supported retraining decisions based on data and behavior changes.
- Communicated analytical findings using visualizations and presentations to support data-driven decision-making.
- Collaborated within **Agile teams** to ensure quality delivery and alignment with business requirements.

Junior Data Scientist | PwC – Hyderabad, India
Jan 2023 – Dec 2023

- Designed and validated **predictive risk models** using regression, classification, and gradient boosted trees across **100+ entities**.
- Framed compliance and risk problems as **data science solutions**, supporting automated and augmented decision-making.
- Built reusable **Python and R pipelines** for data cleaning, transformation, integration, and feature engineering.
- Applied **statistical methods and exploratory analysis** to identify key risk drivers and patterns.
- Evaluated models using **RMSE, R^2, accuracy**, and statistical testing to ensure robustness.
- Deployed predictive models using **Azure Machine Learning**, supporting production use cases.
- Developed **Tableau and Power BI dashboards** to communicate insights to business stakeholders.
- Worked in **Agile, cross-functional teams** with consultants, engineers, and analysts.

Data Analyst | Tata Consultancy Services (TCS) – Remote, India
May 2021 – Dec 2022

- Built analytical and machine learning solutions for customer behavior analysis and operational insights.
- Applied supervised learning techniques following the full data science lifecycle.
- Integrated data from multiple sources using **SQL and NoSQL databases**.
- Supported analytics automation and reporting to improve business decision processes.
- Partnered with cross-functional teams to troubleshoot data issues and improve analytical workflows.

EDUCATION

Master of Science in Computer Science
CGPA – 3.6
University of South Dakota, Vermillion, SD, USA Graduated: May 2025